

Case1:
Sentence 0: The Chteau. de Pirou is a castle in the commune of Pirou in the dpartement of Manche (Normandy) France
Correct answer: contains_administrative_territorial_entity
Baseline: country
DocRE-CLiP: contains_administrative_territorial_entity
Case2:
Sentence 0: The Wigram Baronetcy of Walthamstow House in the County of Essex is a title in the Baronetage of the UNITED KINGDOM . Sentence 3: The second Baronet also represented Wexford Borough in Parliament . Sentence 5: The fourth Baronet was a Lieutenant - General in the army and sat as a Conservative Member of Parliament for South Hampshire and Fareham
Correct answer: legislative_body
Baseline: has_part
DocRE-CLiP: legislative_body
Case3:
Sentence 1: Taking a more electronic music sound than his previous releases TY.O was released in December 2011 by Universal Island Records but for reasons unknown to Cruz its British and American release were held off.] Sentence 3: TY.O features a range of top - twenty and top - thirty singles including “Hang-over” (featuring Florida) “ Troublemaker ” “There She Goes” (sometimes featuring Pitbull) the limited release “World in Our Hands” and “Fast Car” which features on the Special Edition and Fast Hits versions of the album]
Correct answer: Inception
Baseline: publication_date
DocRE-CLiP: publication_date

Table 3: Case study with DocRE-CLiP prediction. Underline text represents entities in the sentence, and purple color represents the DocRE-CLiP prediction.

successfully identifies correct relation *legislative body*. **Case3:** Using pattern recognition, DocRE-CLiP identifies the publication date as a relation. However, the inclusion of the entity context, such as “Troublemaker”, “description”, and “song”, does not contribute significantly to the accuracy of the prediction. Consequently, DocRE-CLiP encounters difficulties in correctly predicting the relation.

Conclusion

This paper introduces DocRE-CLiP, a context-driven approach for document-level relation extraction (DocRE). Our results suggest that integrating diverse context types into the link prediction module enriches relation prediction within the DocRE framework providing interpretability. As future work, researchers can extend our efforts by crafting a versatile model capable of traversing diverse document types, thereby significantly amplifying its aptitude for assimilating knowledge. Furthermore, with conclusive evidence provided in our work as the first step, the document RE and KG link prediction research findings will interchangeably benefit. The code and the models are available at <https://github.com/kracr/document-level-relation-extraction>.

Acknowledgements

We express our sincere gratitude to the Infosys Centre for Artificial Intelligence (CAI) at IIIT-Delhi for their support. RK’s effort has been supported by the U.S. National Library of Medicine (through grant R01LM013240)

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Model	Metric	DocRED	ReDocRED	DWIE
DistMult	Hits@1	0.092	0.061	0.293
	Hits@3	0.104	0.088	0.307
	Hits@10	0.127	0.111	0.334
	MRR	0.105	0.080	0.307
DistMult+context	Hits@1	0.10	0.071	0.297
	Hits@3	0.113	0.093	0.324
	Hits@10	0.112	0.12	0.337
	MRR	0.11	0.12	0.43
Complex	Hits@1	0.092	0.076	0.286
	Hits@3	0.097	0.096	0.296
	Hits@10	0.110	0.104	0.317
	MRR	0.099	0.087	0.297
Complex+context	Hits@1	0.101	0.09	0.31
	Hits@3	0.12	0.10	0.33
	Hits@10	0.15	0.13	0.36
	MRR	0.1	0.98	0.30
R-GCN	Hits@1	0.06	0.033	0.38
	Hits@3	0.09	0.06	0.43
	Hits@10	0.11	0.091	0.45
	MRR	0.0827	0.0532	0.40
R-GCN+context	Hits@1	0.11	0.51	0.67
	Hits@3	0.14	0.61	0.91
	Hits@10	0.23	0.13	0.98
	MRR	1.34	1.13	1.32
KGE-HAKE	Hits@1	0.07	0.05	0.44
	Hits@3	0.103	0.11	0.45
	Hits@10	0.123	0.112	0.47
	MRR	0.09	0.13	0.45
KGE-HAKE+context	Hits@1	0.10	0.08	0.48
	Hits@3	0.156	0.14	0.50
	Hits@10	0.18	0.144	0.51
	MRR	0.136	0.153	0.52

Table 4: Performance of link prediction models.

Query: (IBM research Brazil, parent_organization, ?x)
Answer: IBM research
Explanation: IBM research Brazil, part_of, IBM research
Query: (Piraeus, country, ?x)
Answer: Greece
Explanation: {Piraeus, located_in_the_administrative_territorial_entity, Kiato}
Query: (Quincy, country, ?x)
Answer: United States
Explanation: {Quincy, country, American} {American, country_of_citizenship, America} {America, synonym, United States}

Table 5: Example queries and results on DocRED dataset

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